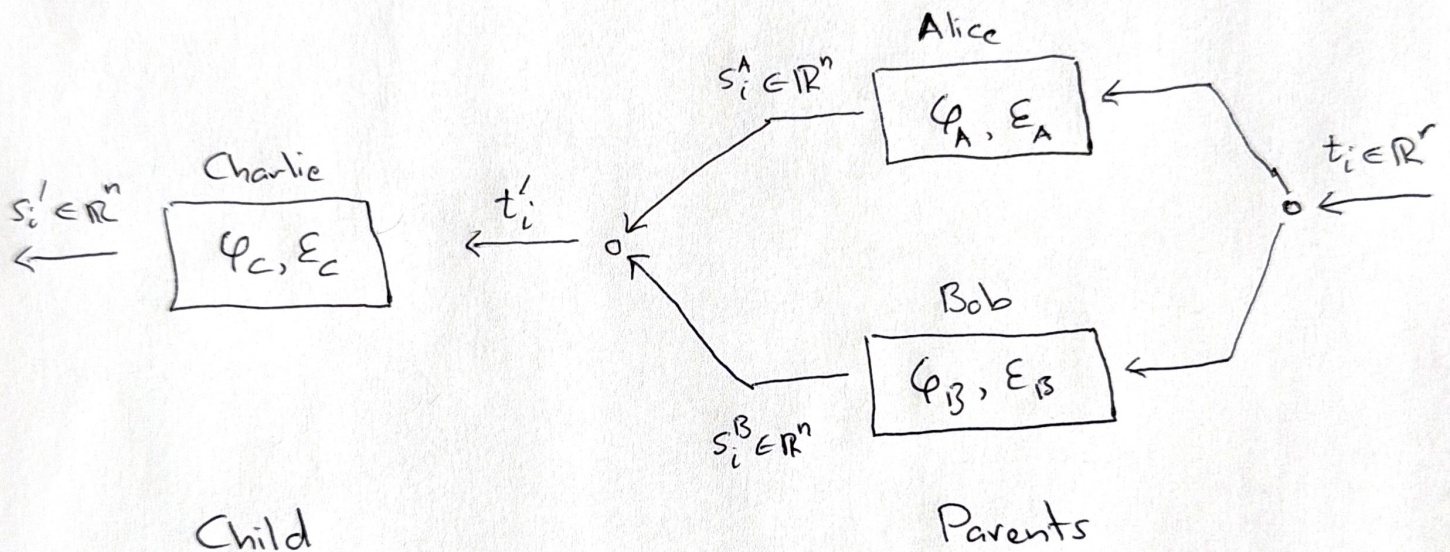


Further Noken genetic operators:

Let $u = (u_1, u_2, \dots, u_n)^T, v = (v_1, v_2, \dots, v_n)^T \in \mathbb{R}^n$

$$u \otimes v = (u_1 v_1, \dots, u_1 v_n, \dots, u_n v_1, \dots, u_n v_n)^T$$

$$u \circ v = (u_1 v_1, u_2 v_2, \dots, u_n v_n)^T$$



$$\varphi_C, E_C = \begin{cases} \varphi_A, E_A & \text{if } E_A < E_B \\ \varphi_B, E_B & \text{otherwise} \end{cases}$$

Kronecker

$$t_i' \triangleq s_i^A \otimes s_i^B$$

Hadamard

$$t_i' \triangleq s_i^A \circ s_i^B$$